

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – Recipe RAG Assistant – An Intelligent Document Q&A Recipe Generator

Domain of Project – Generative AI / Food & Nutrition

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Problem Statement -

Problem Statement :Recipe RAG Assistant

The Challenge

Finding suitable recipes can be difficult when users have **dietary restrictions, limited ingredients, cuisine preferences, and time constraints**. Recipe information is often scattered across different sources and documents. Without **personalized and context-aware guidance**, users spend time searching and manually adapting recipes to their needs.

The Objective

Build an **AI-powered Recipe RAG Assistant** that delivers **personalized recipe recommendations and cooking guidance**. The solution should:

- **Personalized Recipes** – Adapt recipes based on dietary restrictions, ingredients, cuisine, and cooking time.
- **Recipe RAG Retrieval** – Retrieve relevant information from PDF, TXT, and sample recipe documents.
- **Conversational Q&A** – Answer natural-language recipe and cooking questions.
- **Ingredient Substitution** – Suggest alternatives for unavailable ingredients.
- **Nutrition Insights** – Provide estimated calories and nutritional information.
- **Shopping List** – Generate a list of ingredients that are not already available.
- **Source Transparency** – Display the recipe sources used to generate responses.
- **RAG Architecture** – Use **Sentence Transformers + FAISS** for retrieval and **Groq LLM** for response generation.

Proposed Solution -

Proposed Solution: AI-powered Recipe RAG Assistant

The proposed solution is an **AI-powered Recipe RAG Assistant** built using **Streamlit, FAISS, Sentence Transformers, and Groq LLM** to provide personalized, context-aware recipe recommendations. The system addresses challenges in recipe discovery, dietary preferences, ingredient availability, and cooking-time constraints through an intelligent RAG architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant recipe information from uploaded **PDF/TXT files and preloaded recipe documents**. The recipe data is converted into embeddings using **all-MiniLM-L6-v2** and stored in a local **FAISS vector store**, enabling efficient semantic retrieval.

The system provides key capabilities:

Recipe Retrieval retrieves relevant recipe information based on the user's natural-language query.

Personalization Layer adapts recipes according to dietary restrictions, available ingredients, cuisine preferences, and maximum cooking time.

Conversational Q&A allows users to interact with the assistant through a chat interface and receive context-aware recipe recommendations.

Ingredient & Nutrition Assistant suggests ingredient substitutions, provides estimated nutrition information, and identifies missing ingredients.

The solution also generates a **user-friendly structured response** containing step-by-step instructions, substitutions, nutrition information, an automatic shopping list, and the retrieved recipe sources.

Groq's openai/gpt-oss-120b model is used for response generation, while **Sentence Transformers and FAISS** handle semantic embedding and retrieval.

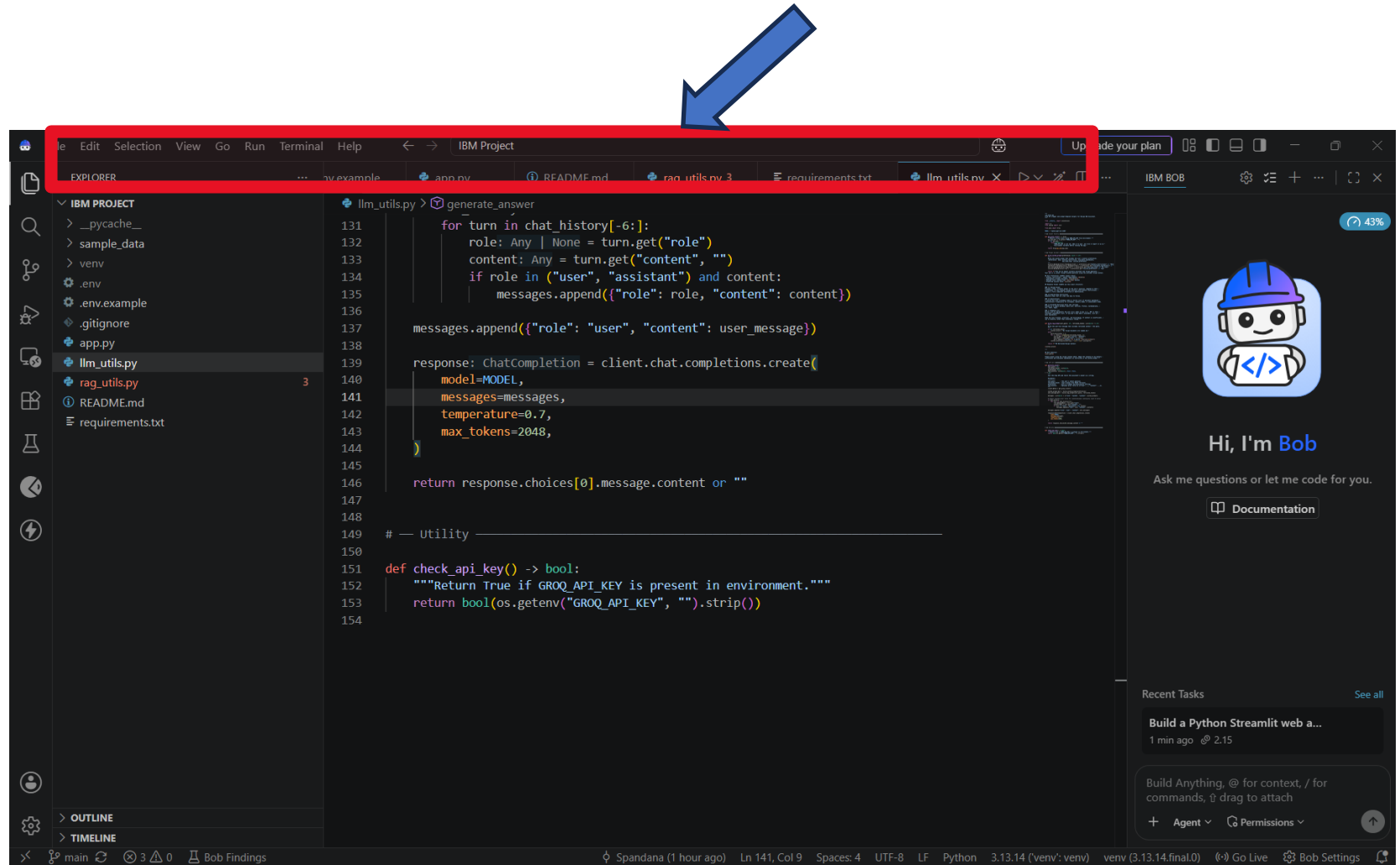
Overall, the solution delivers an **end-to-end intelligent recipe assistant** that combines document-based knowledge retrieval with personalized AI generation to help users discover and prepare recipes according to their individual needs.

Technology Used

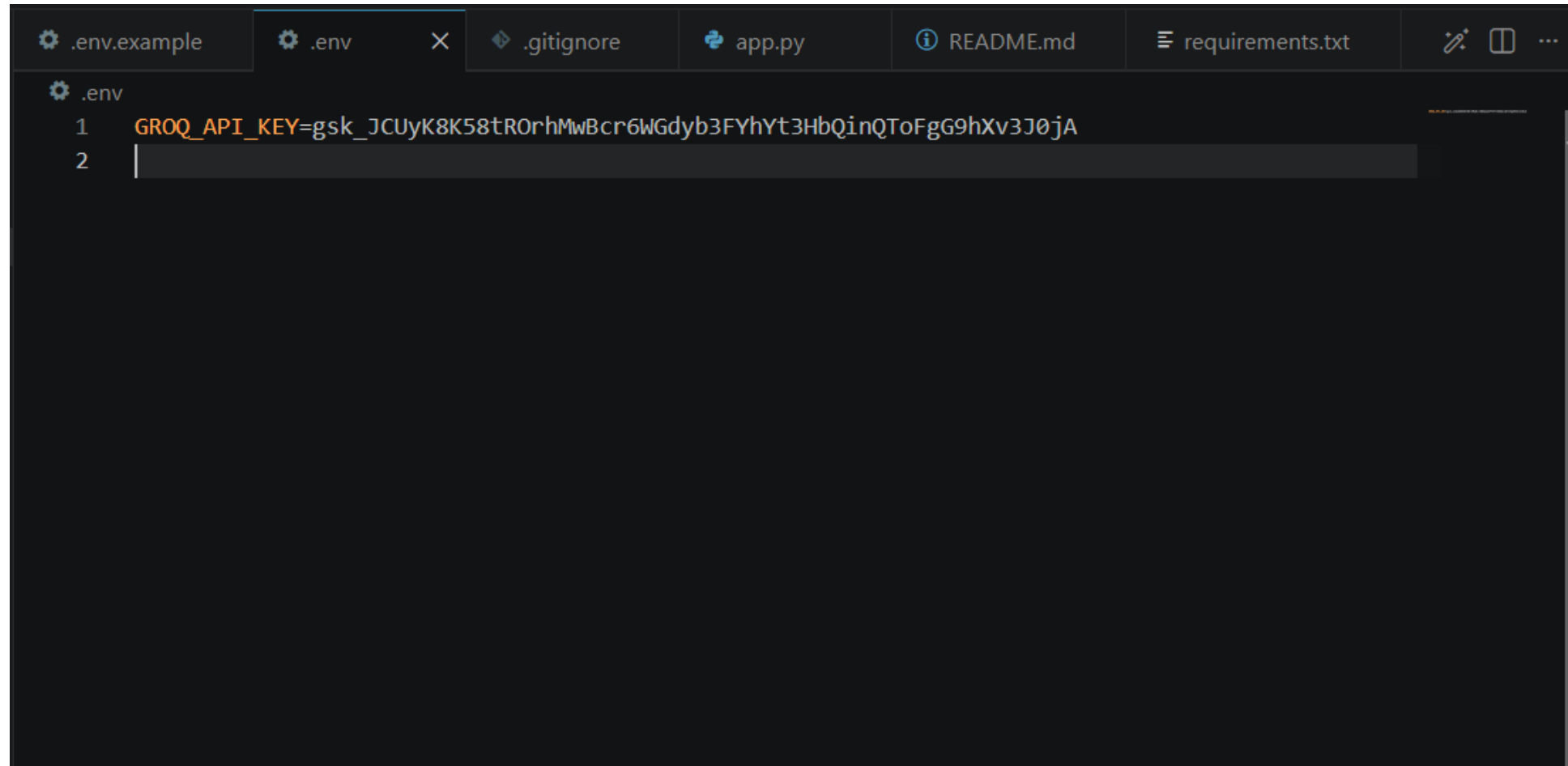
- **Streamlit** – For building the interactive web application and user-friendly recipe chat interface.
- **Groq API** – For accessing the **openai/gpt-oss-120b** model to generate personalized and context-aware recipe responses.
- **Sentence Transformers** – **all-MiniLM-L6-v2** for converting recipe documents and user queries into semantic embeddings.
- **FAISS** – For storing embeddings and performing fast **semantic similarity search** over recipe documents.
- **RAG (Retrieval-Augmented Generation)** – To retrieve relevant recipe information first and then generate grounded, personalized answers.
- **PyPDF** – For extracting recipe content from uploaded PDF documents.
- **Python-dotenv** – For securely loading the **Groq API key** from the `.env` environment file.
- **Git & GitHub** – For version control and project source-code management.

Required Files created using IBM bob -

1. ppp.py
2. requirements.txt
3. sample_data
4. .env
5. rag_utils.py
6. llm_utils.py
7. README.md
8. .env.example



IBM Bob Screenshots:



A screenshot of a code editor interface. The top bar shows several open files: `.env.example`, `.env` (selected), `.gitignore`, `app.py`, `README.md`, and `requirements.txt`. The `.env` file is open, showing two lines of code. Line 1 contains `GROQ_API_KEY=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`. Line 2 is empty. The editor has a dark theme and a sidebar on the right.

```
.env
1 GROQ_API_KEY=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
2
```

```

app.py > ...
1  """
2  app.py – Recipe RAG Assistant
3  Streamlit front-end for a Document Q&A RAG agent that acts as an intelligent
4  recipe generator powered by Groq + FAISS + sentence-transformers.
5  """
6
7  from __future__ import annotations
8
9  import os
10 from pathlib import Path
11
12 import streamlit as st
13 from dotenv import load_dotenv
14
15 # Load .env before any other import that might need the key
16 load_dotenv()
17
18 import rag_utils as rag
19 import llm_utils as llm
20
21 # — Page config —————
22 st.set_page_config(
23     page_title="Recipe RAG Assistant",
24     page_icon="🍳",
25     layout="wide",
26     initial_sidebar_state="expanded",
27 )
28
29 # — Custom CSS —————
30 st.markdown(
31     body="""
32     <style>
33         /* Chat bubbles */
34         .stChatMessage { border-radius: 12px; }
35         /* Sidebar section headers */
36         .sidebar-section { font-weight: 600; margin-top: 0.75rem; color: #3b82d4; }

```


Role of Agentic AI in the solution

Agentic AI enables the Recipe RAG Assistant to function as an intelligent, context-aware system rather than a simple chatbot. It combines **RAG, FAISS, Sentence Transformers, and Groq LLM** to provide personalized recipe guidance.

The system retrieves relevant recipe information from **PDF, TXT, and preloaded recipe documents** using semantic search. **Sentence Transformers** generate embeddings, while **FAISS** identifies the most relevant recipe content for the user's query.

The assistant also considers user preferences such as **dietary restrictions, available ingredients, cuisine preferences, and maximum cooking time** to generate tailored recommendations.

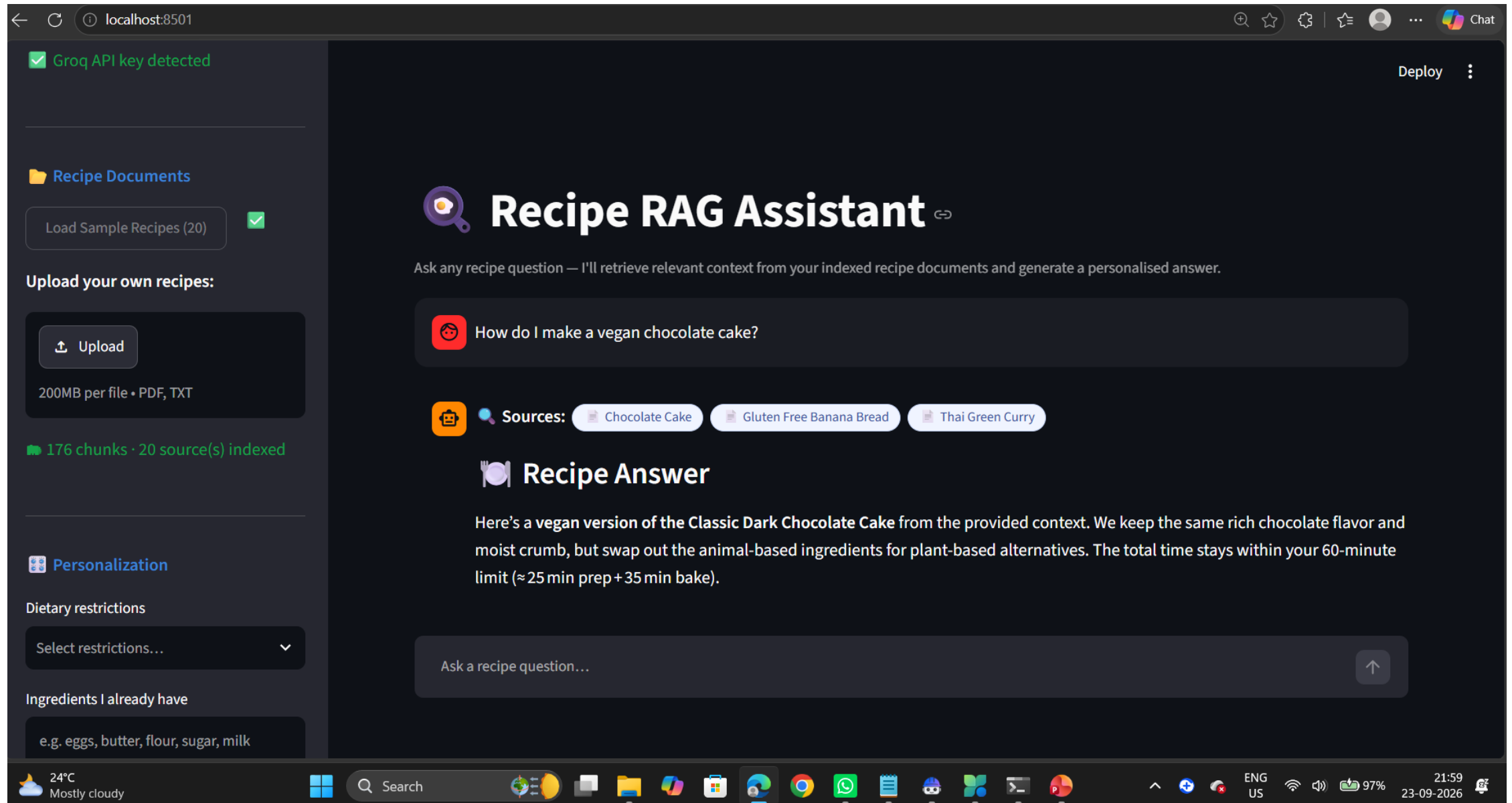
It provides multiple useful outputs including **step-by-step recipes, ingredient substitutions, estimated nutrition information, shopping lists, and recipe sources**.

The system supports **conversational interaction**, allowing users to ask follow-up questions and receive responses based on retrieved recipe context.

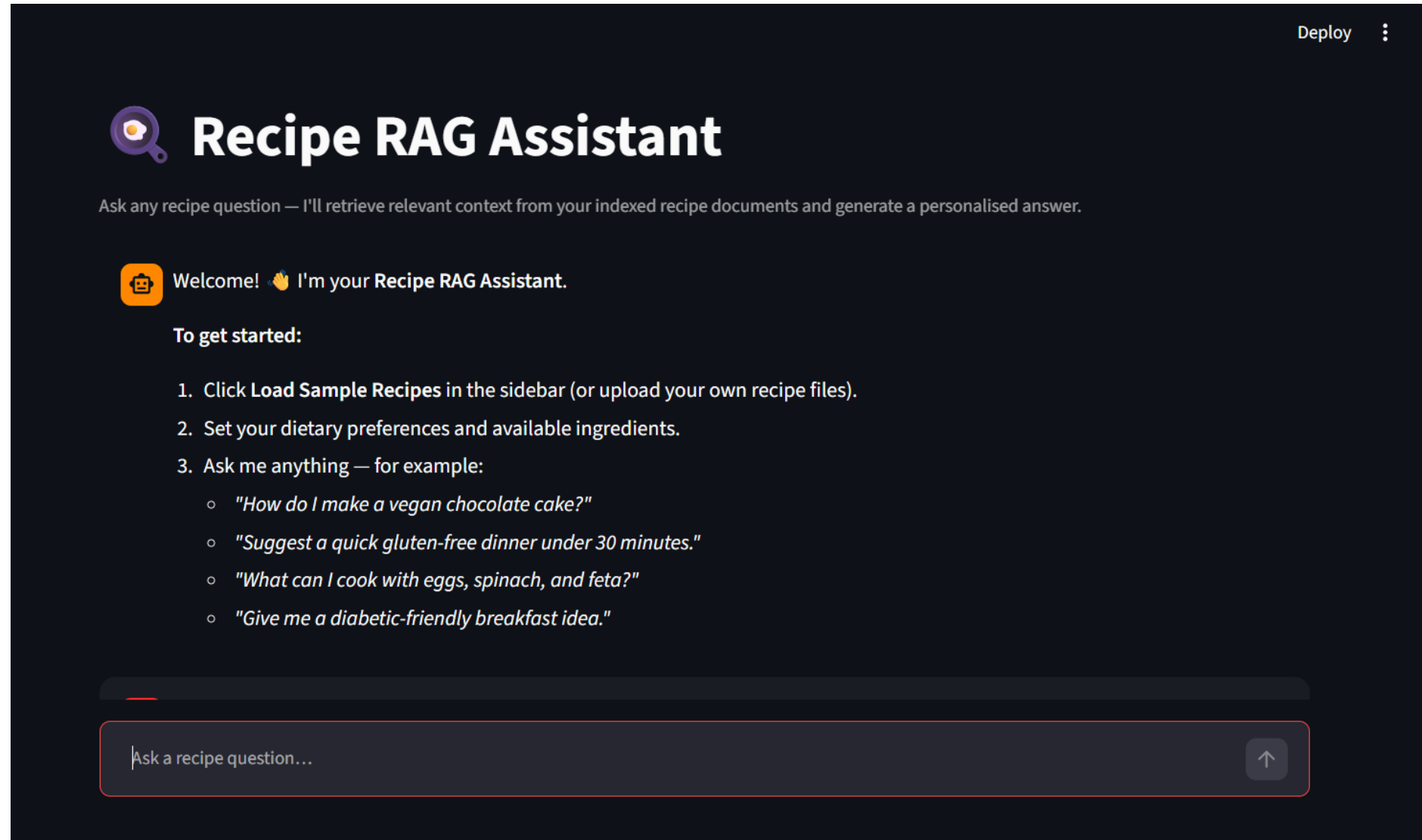
Overall, the use of **RAG and AI-driven personalization** makes the Recipe RAG Assistant a practical and intelligent cooking companion that helps users discover and prepare recipes according to their individual needs.



Project Output Screenshot



Project Output Screenshot



Project Output Screenshot

 "Suggest a quick gluten-free dinner under 30 minutes."

 Sources:

 Gluten Free Banana Bread

 Beef Stew

 Greek Salad

 Diabetic Friendly Oatmeal

 **Recipe Answer** 

A fast, fresh **Greek Salad (Horiatiki)** makes a perfect gluten-free dinner ready in 15 minutes. Pair it with a quick-cook protein like pan-seared chicken breast or grilled shrimp (both can be cooked in 5-10 minutes) to round it out and keep the whole meal under 30 minutes.

 **Step-by-Step Instructions**

- 1. Prep the veggies** – Rinse and cut:
 - 4 medium tomatoes → wedges
 - 1 large cucumber → thick half-moon slices
 - 1 medium green bell pepper → rings
 - 1 small red onion → thin slices
- 2. Make the dressing** (optional but recommended):
 - In a small bowl whisk together 3 Tbsp extra-virgin olive oil, 1 Tbsp lemon juice, ½ tsp dried oregano, a pinch of salt & black pepper.
- 3. Assemble the salad** – In a large bowl combine all the vegetables. Drizzle with the dressing and toss gently.
- 4. Add extras** (optional, still gluten-free):
 - Crumbled feta cheese (≈¼ cup)
 - A handful of Kalamata olives, pitted & halved
- 5. Cook a quick protein** (if desired):

Ask a recipe question...



Novelty and Uniqueness

- **RAG-Based Recipe Assistance** – Uses **FAISS** and **Sentence Transformers** to retrieve relevant recipe information before generating responses.
- **Personalized Recipe Recommendations** – Considers **dietary restrictions, available ingredients, cuisine preferences, and cooking-time limits**.
- **Multi-Source Recipe Support** – Works with **PDF, TXT, and preloaded recipe documents**.
- **Smart Ingredient Substitution** – Suggests suitable alternatives when required ingredients are unavailable.
- **Integrated Nutrition Insights** – Provides **estimated calories and nutritional information** along with recipes.
- **Automatic Shopping List** – Identifies unavailable ingredients and generates a **personalized shopping list**.
- **Source Transparency** – Displays the **recipe sources used** to provide more trustworthy and traceable answers.
- **Conversational Cooking Assistant** – Allows users to ask follow-up questions and receive **context-aware cooking guidance**.

Git Hub Link

- https://github.com/SpandanaRB/recipe-rag-assistant?utm_source=chatgpt.com

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Future Scope

1. Personalized Recipe Recommendations –

The system can provide personalized recipe recommendations based on **dietary restrictions, available ingredients, cuisine preferences, and cooking-time constraints**. This will help users quickly find suitable recipes without manually searching through multiple sources or adapting recipes themselves.

2. Intelligent Recipe Information & Guidance –

The system can use **RAG-based retrieval** to provide relevant recipe information from PDF, TXT, and preloaded recipe documents. It can also offer **ingredient substitutions, nutrition insights, shopping lists, and conversational cooking guidance** based on the user's requirements.

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Certificates Earned:

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Certificates Earned:

Add your IBM BOB Certificate

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Spandana Bathula

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 22 Sep 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.